

MD. ASIBUR RAHMAN BAPPY

Graduate Engineer | VLSI & Semiconductor Manufacturing | Cadence Virtuoso

Dhaka, Bangladesh | 01521564723 | asibur.eee.171@gmail.com | [GitHub](#) | [Portfolio](#)

PROFILE

Graduate Electrical & Electronic Engineer with hands-on experience in full-custom IC design using Cadence Virtuoso and semiconductor layout/verification processes. Designed SRAM cells in 90nm and 45nm technologies with DRC/LVS verification, and received IC Mask Design training at ULKASEMI (16nm). Eager to join a structured Graduate Engineer program with rotations across assembly, test, production, and technology development, growing toward a specialized engineering role.

TECHNICAL SKILLS

Cadence Virtuoso • Cadence Innovus • DRC • LVS • Physical Layout • CMOS Design • SRAM Design • Verilog • Python • Linux (Bash) • MATLAB • C++ • PSpice • Quartus • Proteus • AutoCAD • FreeCAD • CST Studio • Arduino UNO • Raspberry Pi • MS Word / Excel / PowerPoint

KEY VLSI PROJECTS

10T / 6T / 5T / 4T SRAM Cell Design — Project Lead | Academic Project

- Designed transistor-level 10T, 6T, 5T, and 4T SRAM cells across 90nm and 45nm CMOS technology nodes.
- Created schematics and full-custom layouts using Cadence Virtuoso.
- Achieved successful DRC and LVS verification across all variants.
- Compared power, delay, and area trade-offs across the different designs.

Full Adder Using Multiple MOSFET Transistor-Level Logic Styles | Academic Project

- Designed and compared full adder implementations across transistor-level logic styles, optimizing for power, delay, and area.

TRAINING

IC Mask Design Training | ULKASEMI Private Limited

- 5-day intensive training in IC Mask Design using Cadence Virtuoso on a 16nm node; designed/verified layouts, studied advanced short-channel layer rules, and performed DRC/LVS checks; ranked 2nd of 14 participants.

PUBLICATIONS

- Conference Paper: "Design and Performance Analysis of a Highly Sensitive SPR Biosensor for Water Pollution Detection."

OTHER PROJECTS

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| • MATLAB App: GUI-based Differentiation-Integration calculator. | • Student Grade Manager: C++ OOP grade tracker with file handling. |
| • Interactive Websites: Portfolio, photo diary, gallery (HTML/CSS/JS). | • Vibration Meter (IC LM3915): Sensor signal to 10-LED bar graph. |
| • Wireless RC Car with GPS Tracker: Remote control + live location. | • Solar-Based Inverter: DC-to-AC conversion for appliance/grid use. |
| • Adaptive DC Charger: Microcontroller-regulated charging for different battery types. | |

EDUCATION

B.Sc. in Electrical and Electronic Engineering (EEE) | Ahsanullah University of Science and Technology, Dhaka

2021 – 2025 | CGPA: 3.02

Higher Secondary Certificate (HSC) | Government Science College, Dhaka

2020 | GPA: 5.00

RELEVANT COURSEWORK

VLSI & Fabrication Processing, Digital Electronics, Computer Architecture, Electronics I & II, Microcontrollers, Digital Signal Processing

REFERENCES

Dr. Khandakar Md. Ishtiaq — Associate Professor, Department of EEE, Ahsanullah University of Science and Technology (AUST)